Lecture 8
Finite Element Method
Mathematical preliminaries: function spaces, differential operators, functionals, variational calculus

Function spaces

Function Space

- Ω – *open* and *bounded* domain in $\mathbb{R}^n$, where "open" means that the set does not contain its boundary
- $\overline{\Omega}$ – *closure* of Ω, union of Ω and its boundary $\partial\Omega$, i.e., $\overline{\Omega} = \Omega \cup \partial\Omega$
- V – *vector space* over *field* F. For every $\boldsymbol{v}, \boldsymbol{u}, \boldsymbol{w} \in V$ and any $a, b \in F$:
    1. Associativity of addition: $\boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w}$
    2. Commutativity of addition: $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$
    3. Identity element of addition: there is $\mathbf{0} \in V$ such that $\boldsymbol{v} + \mathbf{0} = \boldsymbol{v}$
    4. Inverse element of addition: there is $-\boldsymbol{v} \in V$ such that $\boldsymbol{v} + (-\boldsymbol{v}) = \mathbf{0}$
    5. Compatibility of scalar and field multiplications: $a(b\boldsymbol{v}) = (ab)\boldsymbol{v}$
    6. Identity element of scalar multiplication: $1\boldsymbol{v} = \boldsymbol{v}$, where $1 \in F$
    7. Distributivity of scalar multiplication (1): $a(\boldsymbol{u} + \boldsymbol{v}) = a\boldsymbol{u} + a\boldsymbol{v}$
    8. Distributivity of scalar multiplication (2): $(a + b)\boldsymbol{u} = a\boldsymbol{u} + b\boldsymbol{u}$
- $V(\Omega)$ – *function space*. Given $\Omega \subset \mathbb{R}^n$, $V(\Omega)$ is a space of functions on Ω (by default – vector space) that satisfy some smoothness requirements and possibly boundary conditions on $\partial\Omega$

Normed Space

- Assume $F = \mathbb{R}$
- $\|\boldsymbol{v}\|$ – *norm* of $\boldsymbol{v} \in V(\Omega)$, is an operation $V(\Omega) \to F$ that "measures" the magnitude of the elements of $V(\Omega)$. For any $\boldsymbol{u}, \boldsymbol{v} \in V(\Omega)$ and $a \in F$ a norm must be:
  1. Absolutely homogeneous (scalable): $\|a\boldsymbol{v}\| = |a| \, \|\boldsymbol{v}\|$
  2. Subadditive (triangle inequality): $\|\boldsymbol{u} + \boldsymbol{v}\| \leq \|\boldsymbol{u}\| + \|\boldsymbol{v}\|$
  3. Nonnegative: $\|\boldsymbol{v}\| \geq 0$
  4. Definite (point separating): if $\|\boldsymbol{v}\| = 0$, then $\boldsymbol{v} = \mathbf{0}$
- *Normed vector space* is a pair $(V, \|\cdot\|)$, where V is a vector space and $\|\cdot\|$ is a norm on V
- $(V(\Omega), \|\cdot\|)$ – *normed (function) space*

$L^2(\Omega)$ space

- Let $\Omega = (a, b) \subset \mathbb{R}$
- $L^2(\Omega)$ – space of *square-integrable* functions, i.e.,

$$u \in L^2(\Omega) \;\text{ iff }\; \int_a^b |u(x)|^2\,dx < \infty$$

- $L^2(\Omega)$ is a normed space with L^2-norm defined as

$$\|u\|_2 = \left(\int_a^b |u(x)|^2\,dx\right)^{1/2}$$

- *Inner product* of $u(x)$ and $v(x)$ can be defined as:

$$(u, v)_{L^2} = \langle u, v\rangle_{L^2} = \int_a^b u(x)v(x)\,dx$$

- *Hilbert space* is a normed space, where norm is induced by the inner product. $L^2(\Omega)$ is a Hilbert space, since $\|u\|_2^2 = (u, u)_{L^2}$

$H^1(\Omega)$ space

- $H^1(\Omega)$ – *Sobolev* space of square-integrable functions with square-integrable first derivatives, i.e.,

$$u \in H^1(\Omega) \;\; \text{iff} \;\; u \in L^2(\Omega) \;\; \text{and} \;\; \frac{du}{dx} \in L^2(\Omega)$$

  Hence, $H^1(\Omega) \subset L^2(\Omega)$

- $H^1(\Omega)$ is a normed space with the norm defined as:

$$\|u\|_{H^1} = \left(\int_a^b \left[|u(x)|^2 + \left| \frac{du}{dx} \right|^2 \right] dx \right)^{1/2}$$

- $H^1(\Omega)$ is a Hilbert space since its norm is induced by an inner product:

$$(u, v)_{H^1} = (u, v)_{L^2} + \left(\frac{du}{dx}, \frac{dv}{dx} \right)_{L^2} \quad \Rightarrow \quad \|u\|_{H^1}^2 = (u, u)_{H^1}$$

$C^k(\Omega)$ spaces

- $C(\Omega)$ – space of functions *continuous* on $\overline{\Omega}$
- $C(\Omega) \subset L^2(\Omega)$, since if $u \in C(\Omega)$ and $\overline{\Omega}$ is a *compact* set, then u is *bounded* and, therefore, square-integrable
- With the *uniform* norm:

$$\|u\|_\infty = \max_{x \in \overline{\Omega}} |u(x)|$$

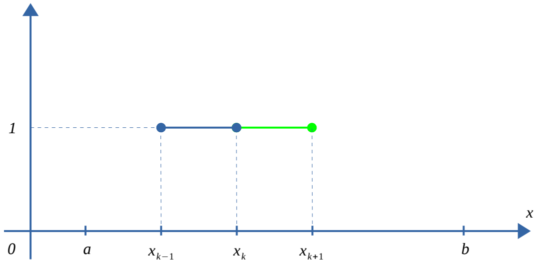
  $C(\Omega)$ becomes a normed space, but *not* a Hilbert space
- $C^1(\Omega)$ – space of functions with *continuous first derivatives* on $\overline{\Omega}$
- $C^2(\Omega)$ – space of functions with *continuous second derivatives* on $\overline{\Omega}$

Piecewise-constant functions

$\Omega = (a, b)$

$$c_k(x) = \begin{cases} 1 & \text{if } \; x \in [x_{k-1}, x_k] \\ 0 & \text{otherwise} \end{cases}$$

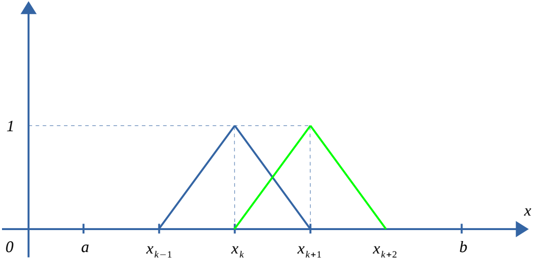
$c_k(x) \in L^2(\Omega)$, $c_k(x) \notin C(\Omega)$, $c_k(x) \notin H^1(\Omega)$



Linear functions: $c_0 + c_1 x$

$$\text{per node: } \ell_k(x) = \begin{cases} \ell_k^-(x) = \frac{x - x_{k-1}}{x_k - x_{k-1}} & \text{if } x_{k-1} \le x \le x_k \\ \ell_k^+(x) = \frac{x_{k+1} - x}{x_{k+1} - x_k} & \text{if } x_k \le x \le x_{k+1} \\ 0 \text{ otherwise} \end{cases}$$

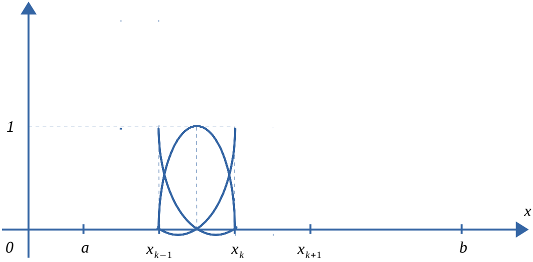
- $\ell_k(x) \in C(\Omega)$, $\ell_k(x) \in H^1(\Omega)$, $\ell_k(x) \in L^2(\Omega)$
- $\ell_k(x) \notin C^1(\Omega)$



Quadratic polynomial functions: $c_0 + c_1 x + c_2 x^2$

per element $x \in [x_{k-1}, x_k]$: $$v(x) = \begin{cases} v_1(x) = \ell^+_{k-1}(2\ell^+_{k-1} - 1) \\ v_2(x) = \ell^-_k(2\ell^-_k - 1) \\ v_3(x) = 4\ell^+_{k-1}\ell^-_k \end{cases}$$

- $v(x) \in C(\Omega)$, $v(x) \in H^1(\Omega)$, $v(x) \in L^2(\Omega)$
- $v(x) \notin C^1(\Omega)$



Function spaces and boundary conditions

- Functions from $C(\Omega)$ satisfying homogeneous Dirichlet BC's:

$$C_0(\Omega) = \{u \in C(\Omega) \,|\, u(a) = u(b) = 0\}$$

- Similarly, we can define $C_0^1(\Omega)$ and $C_0^2(\Omega)$
- Let $\Omega \subset \mathbb{R}^2$, and $\partial\Omega = \partial\Omega_{\mathrm{D}} \cup \partial\Omega_{\mathrm{N}}$, where $u(\boldsymbol{x}) = 0$ if $\boldsymbol{x} \in \partial\Omega_{\mathrm{D}}$. If we say that $u(\boldsymbol{x}) \in C_0^1(\Omega)$, then it means

$$u(\boldsymbol{x}) \in C_0^1(\Omega) = \{u \in C^1(\Omega) \,|\, u(\boldsymbol{x}) = 0, \; \boldsymbol{x} \in \partial\Omega_{\mathrm{D}}\}$$

Note that $\partial\Omega_{\mathrm{N}}$ – Neumann boundary – is not included in the definition of the function space $C_0^1(\Omega)$

Function spaces and boundary conditions

- A function space $V_0(\Omega) = \{u \in V(\Omega) \,|\, u(x) = 0,\ x \in \partial\Omega\}$ with zero Dirichlet boundary conditions is often closed under addition and scalar multiplication, i.e., for any $a, b \in \mathbb{R}$ and $u, v \in V_0(\Omega)$, we have $au(x) + bv(x) \in V_0(\Omega)$
- A function space $V_g(\Omega) = \{u \in V(\Omega) \,|\, u(x) = g(x),\ x \in \partial\Omega\}$ with nonzero Dirichlet boundary conditions is not closed under addition and scalar multiplication, i.e., if $u, v \in V_g(\Omega)$, then, in general, $au(x) + bv(x) \notin V_g(\Omega)$

Differential operators

Differential operators

- To define a *differential operator* one needs a *pair* $(D, V(\Omega))$, where D is an *operation* of differentiation and $V(\Omega)$ is a *function space*. For example, the operator of the second derivative on a function space of twice continuously differentiable functions on $\Omega = (a, b)$ satisfying the BC $u(a) = u(b) = 0$ is defined as

$$\mathcal{L}(u) = \left\{ \frac{d^2 u}{dx^2} \mid u \in C_0^2(\Omega) \right\}$$

- The n-dimensional linear differential operator $\mathcal{L}$ of the order m is a *mapping* (*map*) between two function spaces $\mathcal{L} : V(\Omega) \to W(\Omega)$; $\mathcal{L}(u) = f$, where $u \in V(\Omega)$ and $f \in W(\Omega)$, such that

$$\mathcal{L}(u) = \sum_{|\alpha| \leq m} a_\alpha(x_1, x_2, \ldots, x_n) \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \ldots \partial x_n^{\alpha_n}},$$

where $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$ is a multi-index and $|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$. For example, a two-dimensional second-order differential operator is a mapping of the form:

$$\mathcal{L}(u) = a_{1,1}(x_1, x_2) \frac{\partial^2 u}{\partial x_1^2} + a_{1,2}(x_1, x_2) \frac{\partial^2 u}{\partial x_1 \partial x_2} + a_{2,2}(x_1, x_2) \frac{\partial^2 u}{\partial x_2^2}$$

The role of function space

- Properties of $\mathcal{L}$ depend on the choice of $V(\Omega)$
- Dirichlet boundary conditions at $\partial\Omega_{\mathrm{D}}$ are included in the definition of the operator $\mathcal{L}$ via the function space $V(\Omega)$, e.g., as $V(\Omega) = C_0^1(\Omega)$

Positive differential operators

Similar to positive-definite matrices, a differential operator $\mathcal{L}$ is *positive* if

$$\int_\Omega v\mathcal{L}(v)\, d\Omega > 0, \quad \text{for any} \quad v \in V(\Omega) \setminus \{0\}$$

where $\{0\}$ is the zero element of the function space, e.g., $v(x) = 0$, $x \in \Omega$

Positivity of Laplacian with Dirichlet BC's in 1D

The operator $\mathcal{L}(u) = -u''$ is positive on $C_0^1(a,b)$:

$$\int_\Omega v\mathcal{L}(v)\,d\Omega = -\int_a^b v(x)v''(x)\,dx = \begin{bmatrix} (vv')' = v'v' + vv'' \\ vv'' = (vv')' - v'v' \\ \int_a^b (vv')'\,dx = [vv']_a^b \end{bmatrix}$$

$$= -\left[v(x)v'(x)\right]_a^b + \int_a^b v'v'\,dx = \int_a^b |v'(x)|^2\,dx > 0,$$

since $v \in C_0^1(a,b)$ i.e. $v(x)$ continuous and $v(a) = v(b) = 0$.

Question: can it be that $v(x) = \text{const} \neq 0$, i.e. $v'(x) = 0$, $x \in (a,b)$, and $v(a) = v(b) = 0$?

Positivity of Laplacian with Dirichlet BC's in 2D and 3D

The operator $\mathcal{L}(u) = -\nabla \cdot (k \nabla u)$ is positive on $C_0^1(\Omega)$:

$$\int_\Omega v \mathcal{L}(v)\, d\Omega = -\int_\Omega v \nabla \cdot (k \nabla v)\, d\Omega$$

$$= \begin{bmatrix} \nabla \cdot (v\, k \nabla v) = (\nabla v) \cdot (k \nabla v) + v \nabla \cdot (k \nabla v) \\ v \nabla \cdot (k \nabla v) = \nabla \cdot (v\, k \nabla v) - (\nabla v) \cdot (k \nabla v) \\ \int_\Omega \nabla \cdot \boldsymbol{u}\, d\Omega = \int_{\partial\Omega} \boldsymbol{u} \cdot \boldsymbol{n}\, d\Gamma \end{bmatrix}$$

$$= -\int_{\partial\Omega} (v\, k \nabla v) \cdot \boldsymbol{n}\, d\Gamma + \int_\Omega (\nabla v) \cdot (k \nabla v)\, d\Omega$$

$$= \int_\Omega k \left| \nabla v \right|^2 d\Omega > 0, \quad \boxed{\text{if} \quad k(\boldsymbol{x}) > 0 \;\; \text{and} \;\; v(\boldsymbol{x}) \in C_0^1(\Omega)}$$

Specifically, we need $k(\boldsymbol{x}) > 0$ for all $\boldsymbol{x} \in \Omega$; and $v(\boldsymbol{x}) = 0$, for all $\boldsymbol{x} \in \partial\Omega$

Self-adjoint differential operators

The differential operator $\mathcal{L}$ is *self-adjoint* on $V(\Omega)$ if

$$\int_\Omega v\mathcal{L}(u)\, d\Omega = \int_\Omega u\mathcal{L}(v)\, d\Omega, \quad \text{for all} \quad u, v \in V(\Omega)$$

This is the analogue of the matrix symmetry property.

Self-adjointness of Laplacian with Dirichlet BC's in 1D

The operator $\mathcal{L}(u) = -u''$ is self-adjoint on $C_0^1(a,b)$. Let $u, v \in C_0^1(a,b)$. Then,

$$
\begin{aligned}
\int_\Omega v\mathcal{L}(u)\, d\Omega &= -\int_a^b v(x)u''(x)\, dx \\
&= -\left[v(x)u'(x)\right]_a^b + \int_a^b v'u'\, dx = \int_a^b v'u'\, dx \\
\int_\Omega u\mathcal{L}(v)\, d\Omega &= -\int_a^b u(x)v''(x)\, dx \\
&= -\left[u(x)v'(x)\right]_a^b + \int_a^b u'v'\, dx = \int_a^b v'u'\, dx
\end{aligned}
$$

Self-adjointness of Laplacian with Dirichlet BC's in 2D

Theorem: The operator $\mathcal{L}(u) = -\nabla \cdot (k \nabla u)$ is self-adjoint on $C_0^1(\Omega)$, where $\Omega \subset \mathbb{R}^2$

Proof...

Functionals and variational calculus

Functionals

- *Functional* is a mapping from a function space into the associated field of numbers. In the present case, $L : V(\Omega) \rightarrow \mathbb{R}$
- It can be any mathematical operation that takes a function from $V(\Omega)$ and produces a real number, e.g., integration or taking a maximum

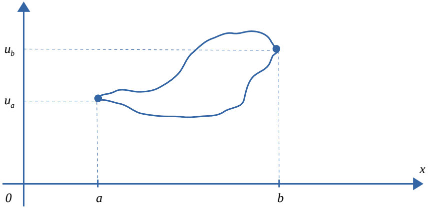
Example: distance on a plane

- Let $V(\Omega)$ be $C(a,b)$ with BC's $u(a) = u_a$ and $u(b) = u_b$ for each $u \in V(\Omega)$. In short

$$V(\Omega) = V(a,b) = \{u(x) \in C(a,b) \mid u(a) = u_a, u(b) = u_b\}$$

  Then, every $u \in V(\Omega)$ is a path between the points (a, u_a) and (b, u_b) in $\mathbb{R}^2$. The function space $V(\Omega)$ is not closed (why?).
- Arc-length of each path is a functional

$$L(u) = \int_a^b \sqrt{1 + (u')^2}\, dx$$



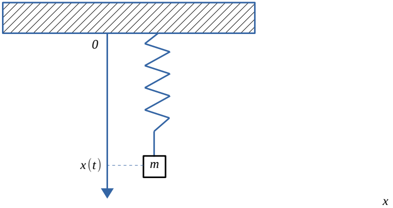
Example: mass on a spring

- We consider $x(t)$ from $C(\Omega)$, where $\Omega = (a, b)$ is an open time interval, and BC's are $x(a) = x_a$, $x(b) = x_b$

$$V(\Omega) = V(a, b) = \{x(t) \in C(a, b) \,|\, x(a) = x_a, x(b) = x_b\}$$

- Kinetic energy is $T = mv^2/2 = m(x')^2/2$. Potential energy is $U = kx^2/2$, where k is the spring coefficient
- *Lagrangian* is $\mathcal{L}(x', x, t) = T - U = m(x')^2/2 - kx^2/2$
- The *action* functional S is

$$S = \int_{t_1}^{t_2} \mathcal{L}(x, x', t)\, dt = \int_a^b \left(\frac{m(x')^2}{2} - \frac{kx^2}{2} \right) dt$$



Minimization of functionals

- *Minimization problem*: find $\tilde{u} \in V(\Omega)$ such that

$$L(\tilde{u}) \leq L(u), \quad \text{for all} \quad u \in V(\Omega)$$

- Introduce an additional *vector* space $V_0(\Omega) \subseteq V(\Omega)$, with homogeneous Dirichlet BC's
  $V_0(\Omega) = \{u \in V(\Omega) \,|\, u(\boldsymbol{x}) = 0, \ \boldsymbol{x} \in \partial\Omega_{\mathrm{D}}\}$
- *Extremum problem*: find $\tilde{u} \in V(\Omega)$ such that the *Gâteaux variation* of L is zero at $\tilde{u}$, i.e.:

$$\frac{d}{d\epsilon} L(\tilde{u} + \epsilon v)\bigg|_{\epsilon=0} = 0, \quad \text{for all} \quad v \in V_0(\Omega)$$

- $V_0(\Omega)$ is called the *space of test functions*
- Rewrite the Gâteaux variation of L so that the du Bois-Reymond Lemma can be applied to derive an (Euler-Lagrange) equation for $\tilde{u}$. Solve the Euler-Lagrange equation on $V(\Omega)$ to find $\tilde{u}$

Fundamental lemma of variational calculus

Lemma (du Bois-Reymond): Assume $\Omega = (a, b)$. Let $V_0(\Omega)$ be a function space of sufficiently smooth functions that vanish on the boundary of Ω. Assume $f(x)$ is continuous on $\overline{\Omega}$, i.e., $f \in C(\Omega)$. Suppose that

$$\int_a^b f(x)v(x)\,dx = 0, \quad \text{for all} \quad v \in V_0(\Omega) \quad (*)$$

Then,

$$f(x) = 0, \quad x \in \overline{\Omega} \quad (**)$$

That is: a *weak* zero of $f(x)$ $(*)$ established for all test functions from $V_0(\Omega)$ is equivalent to the *strong* zero of $f(x)$ $(**)$. Valid also in $2D$ and $3D$.

Example: shortest path in a plane

$\Omega = (a, b)$.
$V(\Omega) = \{u(x) \in C^2(\Omega) \,|\, u(a) = u_a, u(b) = u_b\}$.
$V_0(\Omega) = \{v(x) \in C_0^2(\Omega) \,|\, v(a) = v(b) = 0\}$. Note, $V_0(\Omega)$ is a vector space.
We want to minimize the functional:

$$L(u) = \int_a^b \sqrt{1 + (u')^2}\, dx \quad \text{over} \quad V(\Omega)$$

We look for an extremum, where: $\frac{d}{d\epsilon} L(\tilde{u} + \epsilon v)|_{\epsilon=0} = 0$ for all $v \in V_0(\Omega)$, i.e.,

$$\frac{d}{d\epsilon} L(\tilde{u} + \epsilon v) = \frac{d}{d\epsilon} \int_a^b \sqrt{1 + ((\tilde{u} + \epsilon v)')^2}\, dx$$

$$= \int_a^b \frac{d}{d\epsilon} \sqrt{1 + (\tilde{u}' + \epsilon v')^2}\, dx = \int_a^b \frac{(\tilde{u}' + \epsilon v')v'}{\sqrt{1 + (\tilde{u}' + \epsilon v')^2}}\, dx$$

$$\frac{d}{d\epsilon} L(\tilde{u} + \epsilon v)\Big|_{\epsilon=0} = \int_a^b \frac{\tilde{u}' v'}{\sqrt{1 + (\tilde{u}')^2}}\, dx$$

Shortest path in a plane (continued)

$$\frac{d}{d\epsilon}L(\tilde{u}+\epsilon v)\Big|_{\epsilon=0} = 0, \quad \forall v \in V_0(\Omega)$$

$$\int_a^b \frac{\tilde{u}' v'}{\sqrt{1+(\tilde{u}')^2}}\,dx = \begin{bmatrix} \phi = \frac{\tilde{u}'}{\sqrt{1+(\tilde{u}')^2}}, & d\psi = v'\,dx \\ d\phi = \frac{\tilde{u}''}{(1+(\tilde{u}')^2)^{3/2}}\,dx, & \psi = v \end{bmatrix}$$

$$= \left[\frac{v\tilde{u}'}{\sqrt{1+(\tilde{u}')^2}}\right]_a^b - \int_a^b \frac{v\tilde{u}''}{(1+(\tilde{u}')^2)^{3/2}}\,dx$$

$$= \int_a^b \left[\frac{-\tilde{u}''}{(1+(\tilde{u}')^2)^{3/2}}\right] v\,dx = 0, \quad \forall v \in V_0(\Omega)$$

From du Bois-Reymond's Lemma and since $u \in V(\Omega)$:

$$\frac{-\tilde{u}''}{(1+(\tilde{u}')^2)^{3/2}} = 0, \quad \text{or} \quad \boxed{-\tilde{u}'' = 0,\ x \in (a,b);\ u(a) = u_a,\ u(b) = u_b}$$

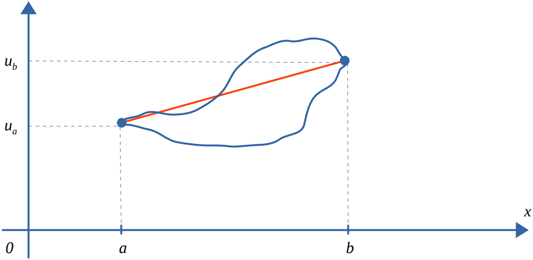
Shortest path in a plane (continued)

Hence, extremum problem for the functional $L(u)$ has been reduced to the boundary-value problem for the differential equation:

$$-\tilde{u}'' = 0, \quad \tilde{u}(a) = u_a, \ \ \tilde{u}(b) = u_b$$

The solution is the straight line between (a, u_a) and (b, u_b):

$$\tilde{u}(x) = u_a + \frac{u_b - u_a}{b - a}(x - b)$$



Lagrange's equation in 1D

- $\Omega = (a, b)$
- $U(\Omega) = \{u(x) \in C^2(\Omega) \,|\, u(a) = u_a, u(b) = u_b\}$
- $V(\Omega) = \{v(x) \in C^2(\Omega) \,|\, v(a) = 0\}$

Minimize

$$S(u) = \int_a^b \mathcal{L}(x, u, u')\, dx \;\; \text{over} \;\; U(\Omega)$$

Additional assumptions:

- $\frac{\partial \mathcal{L}}{\partial u} \in C(\Omega)$
- $\frac{d}{dx}\left(\frac{\partial \mathcal{L}}{\partial u}\right) \in C(\Omega)$
- $V_0(\Omega) = \{v(x) \in C^2(\Omega) \,|\, v(a) = v(b) = 0\}$

Note: $V_0(\Omega) \subset V(\Omega)$.

Lagrange's equation in 1D (continued)

Let $\tilde{u} \in U(\Omega)$, $v \in V(\Omega)$, $\epsilon \in \mathbb{R}$:

$$\begin{aligned}
\frac{d}{d\epsilon} S(\tilde{u} + \epsilon v)\bigg|_{\epsilon=0} &= \frac{d}{d\epsilon} \int_a^b \mathcal{L}(x, \tilde{u} + \epsilon v, (\tilde{u} + \epsilon v)')\, dx \\
&= \int_a^b \frac{d}{d\epsilon} \mathcal{L}(x, \tilde{u} + \epsilon v, \tilde{u}' + \epsilon v')\, dx \\
&= \int_a^b \left[\frac{\partial \mathcal{L}}{\partial \tilde{u}} \frac{d}{d\epsilon}(\tilde{u} + \epsilon v) + \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \frac{d}{d\epsilon}(\tilde{u}' + \epsilon v') \right] dx \\
&= \int_a^b \left[\frac{\partial \mathcal{L}}{\partial \tilde{u}} v + \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v' \right] dx = \int_a^b \frac{\partial \mathcal{L}}{\partial \tilde{u}} v\, dx + \int_a^b \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v'\, dx \\
&= \int_a^b \frac{\partial \mathcal{L}}{\partial \tilde{u}} v\, dx + \left[\frac{\partial \mathcal{L}}{\partial \tilde{u}'} v \right]_a^b - \int_a^b \frac{d}{dx} \left(\frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right) v\, dx
\end{aligned}$$

Lagrange's equation in 1D (continued)

Since $v \in V(\Omega)$, $v(a) = 0$, and $\frac{d}{d\epsilon} S(\tilde{u} + \epsilon v)\big|_{\epsilon=0} = 0 \;\; \Rightarrow$

$$\int_a^b \left[\frac{\partial \mathcal{L}}{\partial \tilde{u}} - \frac{d}{dx}\left(\frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right) \right] v \, dx = - \left. \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v \right|_{x=b}, \;\; \forall v \in V(\Omega) \quad (*)$$

If this holds for all $v \in V(\Omega)$, then it also holds for all $v \in V_0(\Omega)$, since $V_0(\Omega) \subset V(\Omega)$. Recall that $v(b) = 0$ if $v \in V_0(\Omega)$. Further, by du Bois-Reymond's Lemma:

$$\frac{\partial \mathcal{L}}{\partial \tilde{u}} - \frac{d}{dx}\left(\frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right) = 0 \quad (**)$$

On the other hand, if $(**)$ holds, then from $(*)$:

$$- \left. \frac{\partial \mathcal{L}}{\partial \tilde{u}'} v \right|_{x=b} = 0, \;\; \forall v \in V(\Omega)$$

$$\text{i.e. } \left. \frac{\partial \mathcal{L}}{\partial \tilde{u}'} \right|_{x=b} = 0, \quad \text{since } \; v(b) \neq 0$$

Lagrange's equation in 1D (continued)

Hence, the minimization (extremum) problem for the action functional reduces to the Lagrange differential equation:

$$\frac{\partial \mathcal{L}}{\partial \tilde{u}} - \frac{d}{dx}\left(\frac{\partial \mathcal{L}}{\partial \tilde{u}'}\right) = 0, \quad x \in (a, b)$$

with the *essential* boundary condition: $\tilde{u}(a) = u_a$,

and the *natural* boundary condition: $\dfrac{\partial \mathcal{L}}{\partial \tilde{u}'}(b) = 0$

Example: mass on a spring

Identify: $x \to t$, $\tilde{u} \to x$, $\tilde{u}' \to x'$, and the Lagrangian is

$$\mathcal{L}(t, x, x') = \frac{m(x')^2}{2} - \frac{kx^2}{2}$$

The Euler-Lagrange equation is:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial x'}\right) &= 0 \\ -kx - \frac{d}{dt}\left(mx'\right) &= 0 \\ mx'' + kx = 0, \quad t &\in (a, b) \end{aligned}$$

Hence, the Lagrange equation further reduces to a linear second-order ODE ("oscillator" equation).

Lagrangian functional for a self-adjoint operator

Theorem:

- Ω – open bounded domain in $\mathbb{R}^n$, $n = 1, 2, 3$
- $V_0(\Omega)$ – space of functions on Ω that are zero on $\partial\Omega$
- $f \in C(\Omega)$
- $\mathcal{L}$ – positive self-adjoint linear differential operator on $V_0(\Omega)$
- $u_0 \in V_0(\Omega)$ – the unique solution of the differential equation:

$$\mathcal{L}(u) = f$$

Then, u_0 minimizes the functional:

$$L(u) = \int_\Omega \left[\frac{1}{2} u \mathcal{L}(u) - fu\right] d\Omega$$

i.e.

$$L(u_0) \leq L(u), \quad \text{for all} \quad u \in V_0(\Omega)$$

Lagrangian for a self-adjoint operator (proof)

Proof: Let $u, u_0 \in V_0(\Omega)$ and $u \neq u_0$. Also $(u - u_0) \in V_0(\Omega)$ (homogeneous Dirichlet BC's). Since $\mathcal{L}$ is positive on $V_0(\Omega)$,

$$\frac{1}{2}\int_\Omega (u - u_0)\mathcal{L}(u - u_0)\, d\Omega > 0 \quad \text{(positive)}$$

$$\frac{1}{2}\int_\Omega \left[u\mathcal{L}u - u\mathcal{L}u_0 - u_0\mathcal{L}u + u_0\mathcal{L}u_0\right]\, d\Omega \quad \text{(linear)}$$

$$= \frac{1}{2}\int_\Omega \left[u\mathcal{L}u - u\mathcal{L}u_0 - u\mathcal{L}u_0 + u_0\mathcal{L}u_0\right]\, d\Omega \quad \text{(self-adjoint)}$$

$$= \frac{1}{2}\int_\Omega u\mathcal{L}u\, d\Omega - \int_\Omega u\mathcal{L}u_0\, d\Omega + \frac{1}{2}\int_\Omega u_0\mathcal{L}u_0\, d\Omega$$

$$= \frac{1}{2}\int_\Omega u\mathcal{L}u\, d\Omega - \int_\Omega u\mathcal{L}u_0\, d\Omega + \left(1 - \frac{1}{2}\right)\int_\Omega u_0\mathcal{L}u_0\, d\Omega$$

Lagrangian for a self-adjoint operator (proof, continued)

$$
\begin{aligned}
&\frac{1}{2}\int_\Omega (u-u_0)\mathcal{L}(u-u_0)\,d\Omega = \\
&= \frac{1}{2}\int_\Omega u\mathcal{L}u\,d\Omega - \int_\Omega u\mathcal{L}u_0\,d\Omega + \int_\Omega u_0\mathcal{L}u_0\,d\Omega - \frac{1}{2}\int_\Omega u_0\mathcal{L}u_0\,d\Omega \\
&= \frac{1}{2}\int_\Omega u\mathcal{L}u\,d\Omega - \int_\Omega uf\,d\Omega + \int_\Omega u_0 f\,d\Omega - \frac{1}{2}\int_\Omega u_0\mathcal{L}u_0\,d\Omega > 0 \quad (\mathcal{L}u_0 = f) \\
&\frac{1}{2}\int_\Omega u\mathcal{L}u\,d\Omega - \int_\Omega uf\,d\Omega > -\left[\int_\Omega u_0 f\,d\Omega - \frac{1}{2}\int_\Omega u_0\mathcal{L}u_0\,d\Omega\right] \\
&\int_\Omega \left[\frac{1}{2}u_0\mathcal{L}u_0 - u_0 f\right]\,d\Omega < \int_\Omega \left[\frac{1}{2}u\mathcal{L}u - uf\right]\,d\Omega, \quad \text{for all } \; u \in V_0(\Omega)
\end{aligned}
$$

□

From functional to Poisson's equation

- Ω – open bounded domain in $\mathbb{R}^n$, $n = 1, 2, 3$
- $\partial\Omega = \partial\Omega_{\mathrm{D}} \cup \partial\Omega_{\mathrm{N}}$
- $k(\boldsymbol{x}) > 0$, $\boldsymbol{x} \in \Omega$
- $f(\boldsymbol{x}) \in C(\Omega)$
- $g(\boldsymbol{x})$ is a function on $\partial\Omega_{\mathrm{D}}$
- $h(\boldsymbol{x})$ is a continuous function on $\partial\Omega_{\mathrm{N}}$
- $U(\Omega) = \{u \,|\, \nabla \cdot (k\nabla u) \in C(\Omega);\ u(\boldsymbol{x}) = g(\boldsymbol{x}), \boldsymbol{x} \in \partial\Omega_{\mathrm{D}}\}$
- $V(\Omega) = \{v \,|\, \nabla \cdot (k\nabla v) \in C(\Omega);\ v(\boldsymbol{x}) = 0, \boldsymbol{x} \in \partial\Omega_{\mathrm{D}}\}$
- $V_0(\Omega) = \{v \,|\, \nabla \cdot (k\nabla v) \in C(\Omega);\ v(\boldsymbol{x}) = 0, \boldsymbol{x} \in \partial\Omega\} \subset V(\Omega)$

Minimize the functional

$$F(u) = \int_\Omega \left[\frac{1}{2}\nabla u \cdot (k\nabla u) - fu\right] d\Omega + \int_{\partial\Omega_{\mathrm{N}}} hu\, d\Gamma \quad \text{over} \quad U(\Omega)$$

Minimization problem (continued)

Take $u \in U(\Omega)$, $v \in V(\Omega)$, $\epsilon \in \mathbb{R}$, and compute the variation:

$$
\begin{aligned}
& F(u+\epsilon v) = \\
& \int_\Omega \left[\frac{1}{2}\nabla(u+\epsilon v)\cdot(k\nabla(u+\epsilon v) - f(u+\epsilon v)\right]\, d\Omega + \int_{\partial\Omega_\mathrm{N}} h(u+\epsilon v)\, d\Gamma \\
& = \int_\Omega \left[\frac{1}{2}\nabla u\cdot(k\nabla u) + \frac{1}{2}\nabla u\cdot(k\nabla\epsilon v) + \frac{1}{2}\nabla\epsilon v\cdot(k\nabla u)\right. \\
& \qquad \left. + \frac{1}{2}\nabla\epsilon v\cdot(k\nabla\epsilon v) - fu - \epsilon f v\right]\, d\Omega + \int_{\partial\Omega_\mathrm{N}} (hu + \epsilon h v)\, d\Gamma \\
& = \int_\Omega \left[\frac{1}{2}\nabla u\cdot(k\nabla u) - fu\right]\, d\Omega + \int_{\partial\Omega_\mathrm{N}} hu\, d\Gamma \\
& + \epsilon\int_\Omega [\nabla u\cdot(k\nabla v) - fv]\, d\Omega + \epsilon\int_{\partial\Omega_\mathrm{N}} hv\, d\Gamma \\
& + \frac{\epsilon^2}{2}\int_\Omega \nabla v\cdot(k\nabla v)\, d\Omega
\end{aligned}
$$

Minimization problem (continued)

Compute the derivative at $\epsilon = 0$ and transform integral to the form $\int (\dots) v \, d\Omega$

$$\left.\frac{d}{d\epsilon} F(u + \epsilon v)\right|_{\epsilon=0} = 0$$

$$\int_\Omega \left[\nabla u \cdot (k \nabla v) - fv\right] \, d\Omega + \int_{\partial\Omega_\mathrm{N}} hv \, d\Gamma = 0$$

$$\begin{bmatrix} \nabla \cdot (uk\nabla v) = \nabla u \cdot (k\nabla v) + u\nabla \cdot (k \nabla v) \\ \nabla u \cdot (k \nabla v) = \nabla \cdot (vk\nabla u) - v \nabla \cdot (k\nabla u) \\ \int_\Omega \nabla \cdot (vk\nabla u) \, d\Omega = \int_{\partial\Omega} (\boldsymbol{n} \cdot \nabla u) kv \, d\Gamma \end{bmatrix}$$

$$\int_{\partial\Omega} (\boldsymbol{n} \cdot \nabla u) kv \, d\Gamma + \int_\Omega \left[-\nabla \cdot (k\nabla u) - f\right] v \, d\Omega + \int_{\partial\Omega_\mathrm{N}} hv \, d\Gamma = 0$$

$$\int_\Omega \left[-\nabla \cdot (k\nabla u) - f\right] v \, d\Omega + \int_{\partial\Omega_\mathrm{N}} \left[(\boldsymbol{n} \cdot \nabla u) k + h\right] v \, d\Gamma = 0$$

Minimization problem (continued)

$$\int_{\Omega} [-\nabla \cdot (k\nabla u) - f]\, v\, d\Omega + \int_{\partial\Omega_{\mathrm{N}}} [(\boldsymbol{n} \cdot \nabla u)k + h]\, v\, d\Gamma = 0$$

Since this holds for all $v \in V(\Omega)$ it also holds for all $v \in V_0(\Omega)$. Therefore, by du Bois-Reymond's Lemma:

$$-\nabla \cdot (k\nabla u) = f, \quad \boldsymbol{x} \in \Omega \quad (*)$$

Thus, if we choose $u \in U(\Omega)$ to satisfy this equation and the additional boundary condition

$$-(k\nabla u) \cdot \boldsymbol{n} = h, \quad \boldsymbol{x} \in \partial\Omega_{\mathrm{N}}, \quad (**)$$

then we shall minimize the functional $F[u]$ on $U(\Omega)$. The Dirichlet boundary condition $u(\boldsymbol{x}) = g(x)$, $\boldsymbol{x} \in \partial\Omega_{\mathrm{D}}$ is not present in the functional, but is included in $U(\Omega)$ and makes the solution of $(*)$ unique.